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MILITARY SPECIFICATION

MASK ASSEMBLIES, OXYGEN AND SMOKE, FULL FACE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of full face, oxygen and smoke mask assemblies.

1.2 Classification. Mask assemblies shall be of the following types, as specified (see 6.2b):

Type I - Mask, hose assembly (MS90339-3), and connector {MS22016).

Type II - Mask, hose assembly (MS90339-2), and connector (USAF Drawing 55133509).

Type III - Mask, hose assembly (NAVAIR Drawing 3072AS100) and connector NS22016).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (See 6.2c).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, Code SR3, Lakehurst, NJ 08733-5100, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1660

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SPECIFICATIONS

FEDERAL

UU-P-268	-	Paper, Kraft, Wrapping
UU-P-553	-	Paper, Wrapping, Tissue
PPP-B-601	-	Boxes, Wood, Cleated-plywood
PPP-B-636	-	Box, Shipping, Fiberboard
PPP-T-60	-	Tape, Packaging, Waterproof
PPP-T-76	-	Tape, Pressure Sensitive Adhesive, Packaging/Paper (for Carton Sealing)

MILITARY

MIL-P-116	-	Preservation, Methods of
DOD-D-1000	-	Drawings, Engineering and Associated List
MIL-P-5425	-	Plastic, Sheet, Acrylic, Heat Resistant
MIL-C-19246	-	Connector, Oxygen Mask Hose, Type MC-3A
MIL-A-23595/2	-	Amplifier, Audio Frequency AM-4326B/A
MIL-M-26542/6	-	Microphone, M-103/AIC
MIL-M-26542/7	-	Microphone, M-94B/A
MIL-0-27210	-	Oxygen, Aviator’s Breathing, Liquid and Gas

STANDARDS

FEDERAL

FED-STD-595	-	Colors
FED-STD-60l	-	Rubber, Sampling and Testing

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection By Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U-S. Military Property

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MIL-STD-2073-1	-	DOD Materiel Procedures For Development and Application of Packaging Requirements
MS22016		Connector, Oxygen Mask Hose, Type MC-3A
MS90339		Hose Assembly, Oxygen Mask to Connector
MS90340		Insert, Installation and Detail, Full Face oxygen and Smoke Mask

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue,Philadelphia, PA19111-5094.)

* 2.1.2 Other Government documents and drawings. The following other Government documents and drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

3072AS1O0	-	Hose and Communication Assembly

USAF

5583509	-	Connector, Demand Mask to Regulator Tube Internal, Assembly of

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Non-Government publications. The following documents(s) form a part of this document to the extent specified herein. Unless otherwisse specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 256	-	Impact Resistance of Plastics and Electrical Insulating Materials
ASTM D 542	-	Index of Refraction of Transparent Organic Plastics
ASTM D 543	-	Resistance of Plastics to Chemical Reagents

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ASTM D 570	Water Absorption of Plastics
ASTM D 635	Rate of Burning and/or Extent and Time of Burning of Self-Supporting Plastics in a Horizontal Position
ASTM D 638	Tensile Properties of Plastics
ASTM D 648	Deflection Temperature of Plastics Under Flexural Load
ASTM D 695	Compressive Properties of Rigid Plastics
ASTM D 790	Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials
ASTM D 1003	Haze and Luminous Transmittance of Transparent Plastics
ASTM D 1044	Resistance of Transparent Plastics to Surface Abrasion
ASTM D 1149	Rubber Deterioration - Surface Ozone Cracking in a chamber (Flat specimens)
ASTM E 831	Linear Thermal Expansion of Solid Materials By Thermomechanical Analysis
ASTM F 791	Stress Crazing of Transparent Plastics

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from organizations that prepare and distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards),the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The mask assemblies furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable Qualified products List at the time of award of contract (see 4.3 and 6.3). In addition, the retention of the qualification for the mask assembles on the applicable qualified products list shall be dependent on periodic verification of continued compliance with the requirements of this specification (see 4.3 and 4.3.2).

* 3.2 First article. When specified (see 5.2f), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

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3.3 Matertals. Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 Nonmetallic materials. Any nonmetallic material that is adversely affected by continued use with oxygen shall not be used.

3.3.2 Age. Elastomer components, except silicone, shall not be more than 12 months old from the date of manufacture to the date of delivery to any Government service or to any airframe or accessory manufacturer.

3.3.3 Elastomer components. The elastomer components shall be composed of an ozone resistant composition, which shall not bloom, and shall meet the requirements for ozone resistance (see 3.6.11) when tested in 4.7.10.2 and 4.7.11.

3.3.4 Protective treatment. When materials are used in the construction of the masks assemblies that are subject to deterioration when exposed to environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coatings which might crack, chip or scale during normal service life or under extremes of environmental conditions shall not be used.

3.4 Standard parts. MS and AN standard parts shall be used where they suit the purpose. They shall be Identified on the manufacturer's drawings by their part numbers. Parts which are not covered by specifications or military standards, or which are not specifically described herein or on the drawings, shall be completely described on manufacturer's drawings.

3.5 Design and construction. The design and construction of the mask assemblies shall be in accordance with the applicable documents specified in the contract or order (see 6.2b). The mask assembly shall be constructed to permit use of the mask assembly with either a headset (earphones) or a protective helmet without degradation of protection.

3.5.1 Mask.

3.5.1.1 Lens. The lens shall be of one piece construction and shall be - fabricated of an allyl based resin conforming to the physical properties specified in 4.7.10.1. Provisions shall be made for replacement of the lens when required. The mask shall contain provisions for the non-fogging of the lens at an ambient temperature of 50°F and the design of the lens shall be such as to provide for maximum visibility. The optical uniformity of the lens shall conform to the requirements specified in MIL-P-5425.

3.5.1.2 Microphone support. The mask shall contain adequate provisions for the installation of the AM-4326B/A dynamic microphone amplifier conforming to MIL-A-23595/2 and a dynamic microphone conforming to MIL-M-26542/6 or MIL-M-26542/7, as specified by the acquiring activity (see 6.2e). The in-line amplifier for the dynamic microphone shall be placed outside the mask. The location of the support (microphone housing) shall be centrally located such as to provide for maximum fidelity of voice transmittal signals. The Installation of either microphone in this support shall not interfere with normal operation of either the inlet or outlet valves, or with the facial fit of the mask.

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3.5.1.3 Lead-In-tube. The microphone pocket shall contain a 3/16 inch inside diameter lead-in tube for the microphone cable. The tube shall have a blind end to be cut open at the time of insertion of the microphone cable. This opening shall be such that the cable will be evenly centered with a sufficient and even wall thickness around it.

3.5.1.4 Inlet passages.The mask shall contain two inlet ducts, each having a cross-sectional area of not less than 0.2 square inch and each duct shall terminate in an inlet port located in the interior of the mask. The cross-sectional area of each port shall be not less than 0.4 square inch. The other ends of the inlet ducts shall terminate in a common inlet tube exterior to the mask. The inlet tube of the mask shall be furnished with a metalltc insert in accorddance with MS90340. Gas circulation within the mask shall preclude direct flow into the eyes of the user.

3.5.1.5 Exhalation valve assembly. The exhalation valve assembly shall be installed in an exploratory recess. The exhalation va The exhalation valve shall function in conjunction with the inhalation valve to provide pressure compensation and damping.

3.5.1.6 Inhalation valve assembly. The inhalation valve assembly shall be designed to control the oxygen pressure within the mask and respond automatically to changes in the regulator. The valve assembly shall be mounted into place in the inlet tube of the mask ahead of the two inlet ports described in 3.5.1.4. The valve shall provide pressure compensation and damping by means of a compensation tube between the inhalation valve assembly and the exhalation valve assembly.

3.5.2 Hose assembly.

3.5.2.1 Type I. The hose assembly to be furnished with the type I mask assembly shall be in accordance with MS90339-3, except that the color shall be black approximately matching color number 37038 of FED-STD-595.

3.5.2.2 Type II. The hose assembly to be furnished with the type II mask assembly shall be in accordance with MS90339-2, except that the color shall be black approximately matching color number 37038 of FED-STD-595.

3.5.2.3 Type III. The hose assembly to be furnished with the type III mask assembly shall be in accordance with NAVAIR Drawing 3072AS1OO.

3.5.3 Connector.

* 3.5.3.1 Types I and III. The connector to be furnished with the type I and III mask assembly, shall be in accordance with MS22016 and MIL-C-19246, type MC-3A and shall be attached to the free end of the mask hose assembly.

3.5.3.2 Type II. The connector to be furnished with the type II mask assembly, shall be in accordance with USAF Drawing 55B3509 and shall be attached to the free end of the mask hose assembly.

3.6 Performance.

* 3.6.1 Visual examination. When visually examined as specified in 4.7.1.1 through 4.7.1.3, the mask assemblies shall conform to the requirements of this specification.

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3.6.2 Odor. When tested as specified in 4.7.2, the mask shall be judged to have no objectionable odor by at least four of the SIX subjects.

3.6.3 Mask leakage. When tested as specified in 4.7.3, the Inboard leakage shall not exceed 100 milliliters per minute.

3.6.4 Inhalation valve.

3.6.4.1 Low suction inlet performance. The total leakage through the valve inlet portion shall be 15 milliliters per minute or less when tested in accordance with 4.7.4.1.

3.6.4.2 High suction inlet performance. The total leakage through the valve inlet portion shall be 150 milliliters per minute or less when tested in accordance with 4.7.4.2

3.6.4.3 Inspiratory resistance. When tested as specified in 4.7.4.3, the inspiratory resistance of the mask shall not be greater than 1.7 inches of water.

3.6A Exhalation valve.

3.6.5.1 Exhalation performance. Apressure of 20 millimeters of mercury in the face piece and between 15 and 19.9 millimeters of mercury in the inlet hose shall cause the exhalation valve to open when tested in accordance with 4.7.5.1.

3.6.5.2 Exhalation resistance. Exhalation resistance shall not be greater than 4.67 millimeters of-mercury with a flow of 100 liters per minute when tested in accordance with 4.7.5.2.

3.6.6 Subjective tests.

3.6.6.1 Visibility. When tested as specified in 4.7.6.1, the mask lens shall possess the minimum visibility requirements specified below:

Direction	Visibility
Up 0°	50″
up 30°	54°
Up 60°	64°
Side 90°	77°
Side 120°	72°
Side 150°	59°
Down 165°	55°

3.6.6.2 Facial fit. When tested as specified in 4.7.6.2, the mask shall be suitable for all size heads without adjustment other than by means of the head straps. The inside contour of the mask shall be smooth, free from excessive pressure points and other features which might lead to personal discomfort.

3.6.7 Low temperature. When tested as specified in 4.7.7, the mask shall not exhibit any impairment of operation.

3.6.8 High temperature. When tested as specified in 4.7.8, the mask shall not exhibit any impairment of operation.

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3.6.9 High temperature storage. When tested as specified in 4.7.9, the mask shall not be gummy or tacky after the storage period and shall operate satisfactorily.

3.6.10 Physical properties.

3.6. 10.1 Allyl based resins. When tested as specified in 4.7.10.1, the test specimens shall meet the requirements specified in table I.

3.6.10.2 Elastomer components. When tested as specified in 4.7.10.2, the test slabs shall meet the requirements specified in table II.

3.6.11 Ozone-resistance. When tested as specified in 4.7.11, the test slabs shall show no evidence of checking, cracking or other damage under 10X magnification.

3.7 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally Interchangeable. The drawing number requirements of D0D-D-1OOO shall govern changes in the manufacturer's part number.

3.8 color. The color of the elastomer components of the mask assembly shall be black, approximately matching color number 37038 of FED-STD-595.

3.9 Cleanliness. The mask assemblies shall be free of all chemical and mold release agents that might be detrimental to wearer.

3.10 Identification of product. The mask assemblies shall be marked for identification in accordance-with MIL-STD-130, except that the Federal Stock Number shall be omitted from the nameplate. The marking shall include the year and quarter of cure and manufacture of the elastomer components.

3.11 Workmanship. The mask assemblies shall be uniform in quality and shall be free from irregularities, defects or foreign matter which could adversely affect safety, performance, reliability or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any Inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply

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with all requirements of the contract. Sanpling Inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The examination and testing of the masks shall be classified as follows:

a. Qualification Inspection (see 4.3)

b. First article inspection (see 4.4)

c. Quality conformance Inspection (see 4.5)

d. Quality conformance verification inspection (see 4.5.2)

4.3 Qualification inspectton. Qualification Inspection shall consist of all the examinations and tests specified in Table III.

4.3.1 Qualification samples. Qualification samples shall consist of three mask assemblies and six elastomer test slabs, 6 by 6 by 0.075 inches, composed of the identical ozone-resistant composition and cure that was used in manufacturing the masks; and sufficient test specimens to conduct the property tests for the allyl based resins used in the manufcature of the lens. Samples shall be forwarded to a test facility set forth in the letter of authorization to submit samples (see 6.3). The samples shall be plainly identified by securely attached durable tags marked with the following Information:

Sample submitted by (name) (date) for qualification inspection in accordance with the requirements of MIL-M-19417D and number under authorization (reference authorizing letter and number (see 6.3).

4.3.2 Retention of Qualification. The retention of qualification shall consist of periodic verification to determine compliance of the qualified mask assembly with the requirements of this specification. The time and method of periodic verification shall be specified by the activity responsible for the Qualified Products List and shall be included in the Notice of Qualification letter.

4.4 First article inspection. First article inspection shall consist of the examinations and tests specified in Table IV.

4.4.1 First article samples. Unless otherwise specified, as soon as practicable after award of a contract or order, the manufacturer shall submit three masks. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a manufacturer is in continuous production of these masks from contract to contract, submission of further first article samples on the new contract may be waived at the discretion of the acquisition activity (see 6.2f). Approval of the first article samples or the waiving of the first article inspection does not preclude the requirements of submitting to the quality conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the contracting officer (see 6.2g).

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4.4.1.1 First article information. Upon completion of the first article inspection, the Government activity responsible for conducting this inspection program (see 6.2g) shall report to the contracting officer the results of the inspection program, with appropriate recommendations.

4.4.1.2 Disposition of samples. Upon completion of the first article inspection program, the disposition of samples shall be as follows:

a. One approved sample shall be returned to the manufacturer for use in monitoring production.

b. The other samples shall not be returned as they will have been destroyed during the test program. Additionally, these mask assemblies shall not be considered as part of the quantity to be delivered under contract.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in Table V. The sampling and Inspection levels and acceptance criteria-shall conform to MIL-STD-105.

4.5.1 Sampling.

4.5.1. Inspection lot.

4.5.1.1.1 Mask assembly. An Inspection lot shall be expressed in units of mask assemblies made essentially under the same conditions and from the same materials and components. The sample unit shall be one mask assembly.

4.5. 1.1.2 Packaging. An Inspection lot shall be expressed as the number of fully prepared shipping containers, containing mask assembles fully prepared for delivery, made from essentially the same materials and components. The sample unit shall be one shipping container, containing mask assemblies fully prepared for delivery with the exception that it need not be sealed.

4.5.1.2 Sampling for tests and examinations of masks. The sample size, acceptance criteria, tests, and examinations required for the masks shall be as specified in Table V.

4.5.2 Quality conformance verification inspection at an Inspection facility. Upon completion of the tests and examinations specified in Table V, a random sample shall be selected from each lot in accordance with MIL-STD-105, Inpsection Level S-3. The selection of the samples shall be made by the cognizant Government quality assurance representative/specialist. The samples shall be forwarded to the inspection laboratory specified in the acquisition document (see 6.2.h), and subjected to the inspections and tests specified in Table V. The samples shall be assigned by their assigned serial numbers. Additionally, the serial numbers of the units within the lot, represented by the sample units, shall be furnished to the inspection facility. Upon completion of the test and inspection program, the Government activity responsible for conducting the program (see 6.2h) shall report the results of the tests and inspections to the designated inspection and acceptance office specified in the acquisition document. Final acceptance of the lot from which the samples were selected shall be based upon successful completion of the test and inspection program by the cognizant Government quality assurance representative/specialist, applying the applicable outline specified in Table V. Sample units from a rejected lot shall not be resubmitted for tests and examinations without approval of the contracting officer.

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4.6 Test conditions.

4.6.1 Gas. Unless otherwise specified, the gas used in testing the masks shall be oxygen conforming to MIL-0-27210, type I.

4.6.2 Temperature and pressure. Unless otherwise specified, inspections shall be conducted at local ambient temperatures and barometric pressure. The temperature and barometric pressure shall be recorded at the time of Inspection. This information shall be available for computation of test data, where required, to normal temperature and pressure (NTP) conditions. NTP conditions are 70°F and 29.92 inches of mercury. Test instruments shall be calibrated or adjusted according to their required usage in conducting individual tests.

4.7 Inspection methods.

4.7.1 Visual examination.

4.7.1.1 Mask assemblies. Every mask assembly shall be examined visually (for critical defects) to determine conformance to this specification and applicable drawings. The classification of defects, table VI, shall be used to classify the defects found.

4.7.1.2 Dimensions. The mask assembly shall be checked dimensionally to determine conformance to applicable standards.

4.7.1.3 Packaging. Each of the fully prepared shipping containers containing mask assemblies selected as a sample unit from the lot shall be examined to determine that the packaging, packing, and marking conform to this specification. The classification of defects table VII, shall be used to classify the defects found.

4.7.2 Odor. The mask shall be worn by six subjects in a manner that will prevent exterior odors from influencing the test. The mask shall pass the requirements specified in 3.6.2.

4.7.3. Mask leakage. The mask assembly shall be fitted to a suitable facsimile of a head simulating service conditions and adjusted so there is no leakage around the face seal. The free end of the hose shall be plugged. With an internal suction of 13 millimeters of mercury applied at standard atmospheric conditions, the mask shall conform to the requirements of 3.6.3.

4.7.4 Inhalation valve.

4.7.4.1 Low suction inlet performance. The inlet valve shall be completely dry throughout this test. The inlet valve shall be inspected for imperfect seating of the flapper on the seat. The valve shall be inspected for leaks around the valve seat by applying and maintaining a suction of 0.19 millimeters of mercury at the free end of the inlet hose. The inlet valve shall conform to the requirements of 3.6.4.1.

4.7.4.2 High suction inlet performance. This test shall conform to 4.7.4.1 except that a suction of 22.4 millimeters of mercury shall be used. The inlet valve shall conform to the requirements of 3.6.4.2.

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4.7.4.3 Inspiratory resistance. The mask shall be fitted to a suitable facsimile of a head simulating service condition. The connector assembly shall not be attached. A flow of 70 liters per minute shall be applied to the valve and maintained for a period of 2 minutes. The mask shall pass the requirements specified in 3.6.4.3.

4.7.5 Exhalation valve.

4.7.5.1 Exhalation performance. The mask assembly shall be placed on a suitable test stand and all suction and pressure from the previous test exhausted. A pressure of 20 millimeters of mercury shall be applied at the mask hose. The pressure at the mask hose shall then be gradually reduced, causing the inlet valve to close and causing a pressure differential to be applied across the exhalation valve. The exhalation valve shall conform to the requirements of 3.6.5.1.

4.7.5.2 Exhalation resistance. The mask assembly shall be placed on a suitable test stand. With an flow of 100 liters per minute, the mask assembly shall conform to the requirements of 3.6.5.2.

4.7.6 Subjective tests. The following subjective tests shall be conducted simultaneously utilizing six subjects.

4.7.6.1 Visibility. Visibility requirements shall be determined by using the method shown on figure 1. The mask lens shall pass the requirements specified in 3.6.6.1.

4.7.6.2 Facial fit. The mask shall be evaluated subjectively for facial fit and shall pass the requirements specified in 3.6.6.2.

4.7.7 Low temperature. The mask assembly shall be conditioned at -65″ ± 2°F for two hours and then returned to -40° ± 2°F. While at this temperature, the mask assembly shall be evaluated subjectively or simulated mechanically for pliability and suitability for operation. The mask assembly shall pass the requirements specified in 3.6.7.

4.7.8 High temperature. The mask assembly shall be conditioned at 160° ± 2°F for two hours. While at this temperature, the mask assembly shall be evaluated subjectively or simulated mechanically for pliability and suitability for operation. The mask assembly shall pass the requirements specified in 3.6.8. The mask assembly shall then be subjected to and pass the odor test.

4.7.9 High temperature storage. The mask, exclusive of plastic parts, shall be placed in an atmosphere maintained at a temperature of not less than + 212°F for a period of not less than 72 hours. The mask shall pass the requirements specified in 3.6.9. The mask shall then be subjected to and pass the odor test.

4.7.10 Physical properties.

4.7.10.1 Allyl based resins. Three allyl based resin specimens (size as determined from the applicable test methods) shall be tested in accordance with the methods specified in Table I. The specimens shall pass the requirements specified in 3.6.10.1.

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4.7.10.2 Elastomer components. Three of the SIX elastomer test slabs submitted for qualification tests (4.3.1) shall be tested in accordance with the specified methods of FED-STD-601 (see table 11). The test slabs shall pass the requirements specified in 3.6.10.2.

4.7.11 Ozone resistance test (elastomer components). Three of the six elastomer test slabs submitted for qualification inspections shall be tested for ozone-resistance. The test apparatus shall be in accordance with ASTM D 1149. The test slabs shall be elongated 20 percent, placed in an ozone-free atmosphere for 24 hours, then placed in the ozone chamber. The chamber shall be adjusted to 100° ± 2°F and to give an exposure of ozone concentration of 120 +10 parts by volume of ozone per million parts by volume of air. The air-ozone velocity in the chamber shall be at least 2 feet per second. The material shall be exposed to these conditions for 60 minutes. The test slabs shall be examined under a 10X magnification and shall pass the requirements specified in 3.6.11.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, as specified (see 6.2i).

5.1.1 Level A.

5.1.1.1 cleaning. Mask assemblies shall be thoroughly cleaned for oxygen service in accordance with industry practice. Petroleum and other flammable solvents shall not be used.

5.1.1.2 Drying. Mask assemblies shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservation application. Preservatives shall not be used.

5.1.1.4 Unit packaging. Unless othewise specified by the acquisition activity, each mask assembly shall be individually unit packed in accordance with MIL-P-116, Method 1A-15. The connectors of the mask assembly shall be closed and cushioned. Mask assembly shall be cushioned within a PPP-B-636 unit container by use of tissue wrapping paper in accordance with UU-P-553 or equal material shall be free from all lint, dust, oil or grease. The PPP-B-636 unit container shall be overwrapped with Kraft paper in accordance with UU-P-268.

5.1.1.5 Intermediate packaging. No more than six mask assemblies, unit packed as specified in 5.1.1.4, shall be placed in Intermediate containers conforming to PPP-B-636. Seams, joints and closures sshall be taped with material conforming to PPP-T-60, Class I or PPP-T-76.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2i).

5.2.1 Level A. Mask assemblies, packaged as described in 5.1, shall be packed in a wood-cleated plywood box in accordance with PPP-6-601, overseas type. The container shall be strapped in accordance with the appendix of the container specification.

5.2.2 Level B. Mask assemblies, packaged as described in 5.1, shall be packed in a fiberboard box in accordance with PPP-B-636, Class weather resistant, variety DW. Seams, joints and closures shall be taped with material conforming to PPP-T-60, Class I or PPP-T-76.

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5.2.3 Level C. The level C packing of mask assemblies shall conform to MIL-STD-2073-1.

5.3 Marking. Unit, intermediate and shipping containers shall be marked in accordance with the requirements of MIL-STD-129 (see 6.2j).

5.3.1 Special marking. The following precautionary marking shall appear on each package:

CAUTION

DO NOT ALLOW CONTAMINANTS OF ANY KIND
TO BE USED ON OR ABOUT THE OXYGEN MASKS.

5.3.2 Additional marking. In addition, the following shall be marked on all packages and containers:

WHENEVER POSSIBLE, STORE MASKS AT TEMPERATURE OF 32° to 80°F

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The combination oxygen and smoke mask assemblies covered by this specification are for use in dispensing gaseous oxygen from demand regulators to aircrew members and to protect flight crews from the harmful effect of smoke, carbon monoxide, or other incapacitating gases.

6.2. Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification (including amendments).

b. Applicable type of mask assembly required (see 1.2).

c. Issue of DODISS to be citied in the solicitation , and if required, the specific issue of individual documents referenced (see 2.1.1. and 2.2.)

d. Applicable drawing and part number of mask required (see 3.5).

e. Dynamic microphone required (see 3.5.1.2).

f. Whether first article inspection is waived (see 4.4.1)

g. Name and address of the first article inspection test facility (see 4.4.1) and the name and address of the Government activity responsible for conducting the first article inspection program (see 4.4.1.1 and 6.5).

h. Name and address of the quality conformance verification inspection laboratory (see 4.5.2.1) and the name and address of the Government activity responsible for conducting the quality conformance verification inspection program (see 4.5.2.1).

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i. Selection of applicable levels of preservation, packaging and packing required (see 5.1 and 5.2).

j. Whether any special markings are required (see 5.3).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of a ward of contract, qualified for includion in Qualified Products List No.QPL-19417 whether or not such products have actually been so listed by that date. The date. The attention of the contractors is called to these requirements, and manufactures are urged to arrange to have the products that they propose to of fer to the Federal Government tested for qualification in order that they might be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Air Systems Command, Department of the Navy, Washington,DC 20361; however, authorization for qualification of products shall be obtained from the Commanding Officer, Naval Air Warfare Center Aircraft Division Warminster, (Code 6031), Warminster, PA 18974. Prior to submission of the samples for qualification inspection, the manufacturer shall submitt a request to the Naval Air Warfare Center Aircraft Dlvision Warminster (Code 6031) indicating a date on which the samples can be forwarded and also request anauthorization number to accompany the samples.

6.4 First article. When a first article inspection is required, the Item(s) should be a first article sample. The contracting officer should include specific instructions in acquisition documents, regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, must furnish evidence with the bid prior to Government approval that the item is presently appropriate for the pending contract.

6.5 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of-the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.4.1.1	DI -NDTI-80809A	Test/Inspection Report	Use contontractor format

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.6 Metrication. Any metrication shall conform to the practices of FED-STD-376. If metrication is utilized, the mask assemblies shall remain in conformance with all requirements of this specification, within specified limits.

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*

6.7 Subject term (key word) listing.

Connector, mask
Gas, breathing
Hose assembly
Lens mask
Life support
Protection, Incapacitating gases
Protection, smoke

6.8 Changes from previous issue. The margins of this specification are marked with asterisks (or vertical lines) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contracters are cautioned to evaluate the requirements of this document based on the entire content Irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - AV
Navy - AS
Air Force - 99

Review Activity:
Air Force - 11

Preparing Activity
Navy - AS
(Project No. 1660-0517)

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TABLE I. Physical property tests, allyl based resins.

Property	Method	Requirement
Tensile strength, psi, min.	ASTM D 638 1/	8000
Compressive strength, psi, min.	ASTM D 695	20,000
Flexural strength, psi, min.	ASTM D 790 2/	13,500
Izod impact strength, ft. lb per inch of notch, min.	ASTM D 256 3/	0.30
Visible light transmission, percent, min.	ASTM D 1003 4/	92
Index of refraction	ASTM D 542	1.50 ± 0.01
Linear thermal expansion, coefficient, maximum per °C	ASTM E 831	9.7×10^{-5}
Deflection temperature @ 0.010 inch deflection, °C, min.	ASTM D 648 5/	67
Rate of burning, maximum inches/minute	ASTM D 635	0.9
Water absorption after 24 hours, percent maximum	ASTM D 570	0.30
Surface abrasion, X methacrylate	ASTM D 1044	20
Acetone, absorption, percent	ASTM D 543	1.2
Crazing resistance under stress	ASTM F 791	No cracks or crazing

1/ Type I specimen.
2/ Method I.
3/ Method A.
4/ Procedure A.
5/ Procedure 6(a).

TABLE II. Physical property tests, elastomer components.

Property	Method 1/	Requirement
Original		
Modulus (100% Elongation), psi, min.	4131	100
Tear Resistance (Type B die), lb/in, min.	4211	100
Tensile Strength, psi, min.	4111	1000
Ultimate Elongation, percent minimum	4121	400
Shore Durometer Hardness	3021	50 ± 5
Tension Set, percent maximum	4411	10
Accelerated Aging at 212 ± 5°F for 72 hours		
Tensile Strength, percent loss, maximum	7221	15
Ultimate Elongation, percent loss, maximum	7221	15
Compression Set (30 percent deflection), percent maximum	3311	35

1/ Method as specified in FED-STD-601.

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TABLE III. Qualification Inspection.

Inspection	Requirement paragraph(s)	Test method Paragraph(s)
Visual examination	3.6.1	4.7.1.1
Dimensional examination	3.6.1	4.7.1.2
Odor	3.6.2	4.7.2
Mask leakage	3.6.3	4.7.3
Low suction inlet performance	3.6.4.1	4.7.4.1
High suction inlet performance	3.6.4.2	4.7.4.2
Inspiratory resistance	3.6.4.3	4.7.4.3
Exhalation performance	3.6.5.1	4.7.5.1
Exhalation resistance	3.6.5.2	4.7.5.2
Visibility	3.6.6.1	4.7.6.1
Facial fit	3.6.6.2	4.7.6.2
Low temperature	3.6.7	4.7.7
High temperature	3.6.8	4.7.8
High temperature storage	3.6.9	4.7.9
Physical properties - Allyl based resins	3.6.10.1	4.7.10.1
Physical properties - Elastomer components	3.6.10.2	4.7.10.2
Ozone resistance - Elastomer components	3.6.11	4.7.11

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TABLE IV. First article inspection.

Inspection	Requirement paragraph(s)	Test method Paragraph(s)
Visual examination	3.6.1	4.7.1.1
Dimensional examination	3.6.1	4.7.1.2
Odor	3.6.2	4.7.2
Mask leakage	3.6.3	4.7.3
Low suction inlet performance	3.6.4.1	4.7.4.1
High suction performance	3.6.4.2	4.7.4.2
Inspiratory resistance	3.6.4.3	4.7.4.3
Exhalation performance	3.6.5.1	4.7.5.1
Exhalation resistance	3.6.5.2	4.7.5.2
Visibility	3.6.6.1	4.7.6.1
Facial fit	3.6.6.2	4.7.6.2
Physical properties - Allyl based resins	3.6.10.1	4.7.10.1
Physical properties - Elastomer components	3.6.10.2	4.7.10.2
Ozone resistance	3.6.11	4.7.11

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TABLE V. Quality conformance inspection.

INSPECTION	METHOD	SAMPLE SIZE	ACCEPTANCE CRITERIA
Visual examination (See classification of defects.)	4.7.1.1	Every mask for critical defects. Inspection Level II 1/ for minor defects.	Reject all units with any critical defects. An acceptable quality level of 2.5 defects per hundred units for minor defects.
Dimensions	4.7.1.2	Inspection Level S-2 1/	Acceptance number zero, rejection number 1.
Odor	4.7.2	Inspection Level S-2 1/	Acceptance number zero, rejection number 1.
Mask leakage	4.7.3	Every mask	Reject all defective units.
Low suction inlet performance	4.7.4.1	Every mask	Reject all defective units.
High suction inlet performance	4.7.4.2	Every mask	Reject all defective units
Inspiratory resistance	4.7.4.3	Every mask	Reject all defective units.
Exhalation performance	4.7.5.1	Every mask	Reject all defective units
Exhalation resistance	4.7.5.2	Every mask	Reject all defective units
Visibility	4.7.6.1	Inspection Level S-2 1/	Acceptance number zero, rejection number 1.
Packaging	4.7.1.3	Inspection Level S-2 1/	Total acceptable quality level of 4.0 percent defective.

1/ The sample size shall be based only on the applicable sample size code letter corresponding to the specified inspection level of MIL-STD-105.

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TABLE VI. Classification of defects for visual examination of the mask assembly.

CRITICAL	MINOR
1. Material imperfections - foreign matter embedded.	201. Marking - missing, insufficient, incorrect, illegible, or not permanent.
2. Surface - unclean, rough, misaligned, or containing cracks, nicks, or other flaws.	202. Color not as specified.
3. Any component missing, malformed, fractured, or otherwise damaged.	
4. Any component loose or otherwise not securely retained.	
5. Incorrect assembling or improper positioning of components.	
6. Any functioning part that works with difficulty.	
7. Faulty workmanship or other irregularities.	

TABLE VII. List of defects for packaging.

ITEM	DEFECTS
Exterior and interior markings	Missing, incorrect, incomplete, illegible; of improper size, location, sequence; or method of application; markings not the same on the interior and exterior containers.
Packaging and Packing materials	Any non-conforming component; any component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of the components such as imcomplete closure of the unit package, intermediate package, container flaps, loose strapping, etc.; bulging or distortion of the containers.
Exterior and interior weight or content	Number per container is more or less than required; gross or net weight exceeds the requirements.

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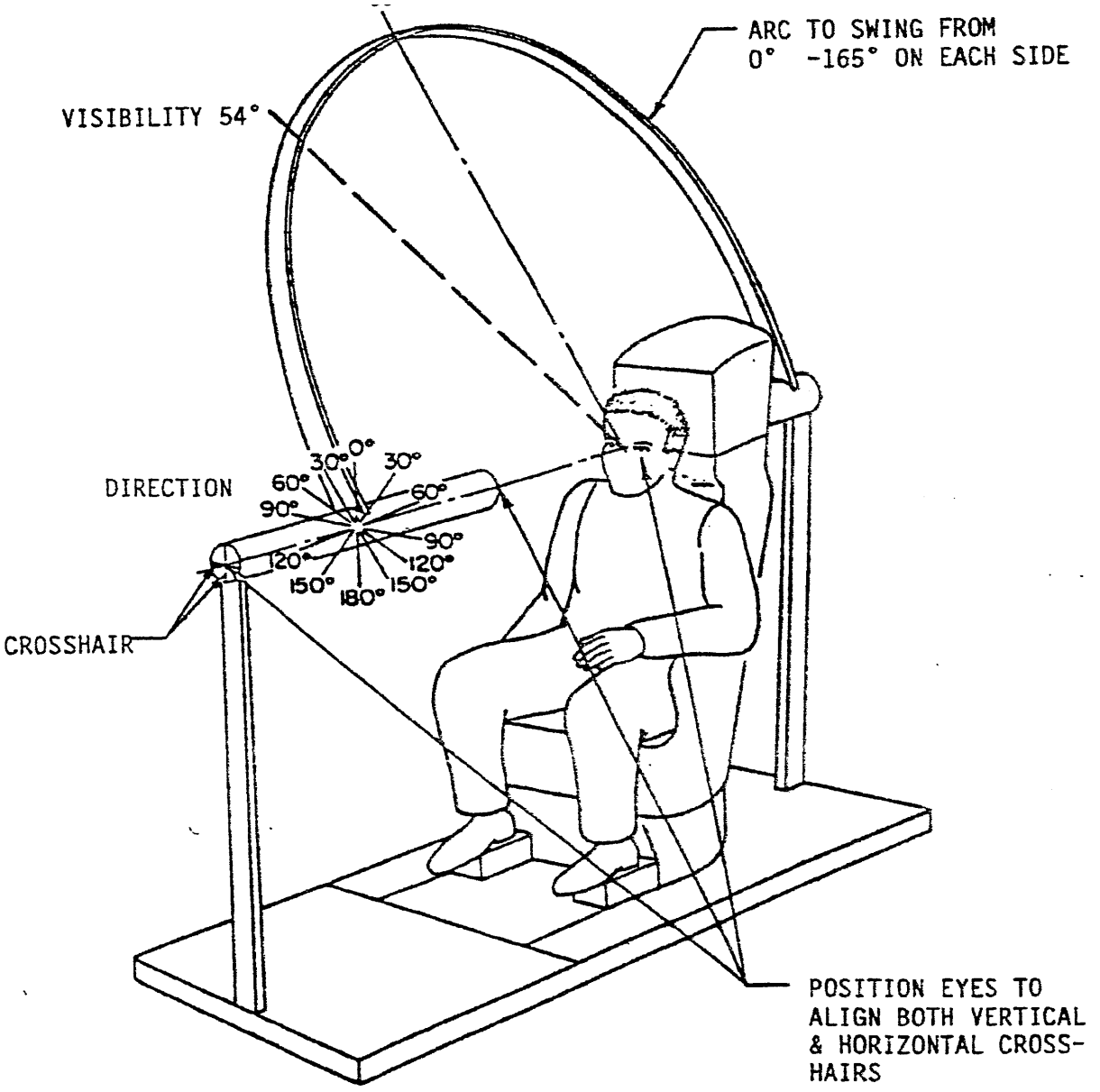


Figure 1. Method of determining visibility requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-19417D	92/09/29

3. DOCUMENT TITLE
MASK ASSEMBLIES, OXYGEN AND SMOKE, FULL FACE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDING OFFICER, NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION LAKEHURST SYSTEMS REQUIREMENTS DEPARTMENT	b. TELEPHONE (Include Area Code) (1) Commercial: (908) 323-7488 (2) AUTOVON: 624-7488
c. ADDRESS (Include Zip Code) CODE SR3 LAKEHURST, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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